

Daniel Gause

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Education

B.A. Physics and Mathematics

September 2016 – May 2020

Middlebury College

- GPA 3.5, *cum laude*
- Physics thesis: Structural Analysis of Potential Dual Quasar Pairs
- Mathematics thesis: An Introduction to Large Deviation Theory

Research Experience

Postbaccalaureate Researcher

September 2026 – Present

National Radio Astronomy Observatory — Advisor: Adele Plunkett

- Conducting a multiwavelength study of protostellar outflows using archival data from ALMA, the VLA, and the Hubble Space Telescope
- Continuing development of MANNA, with a deployment on the NOIRLab Astro Data Lab in internal testing

Research Intern

May 2026 – August 2026

NSF-Simons AI Institute for Cosmic Origins (CosmicAI) — National Radio Astronomy Observatory

- Built [MANNA](#), an open-source MCP server that exposes the NOIRLab and NRAO/ALMA data archives to natural-language queries through IVOA standards (TAP, SIA, Cone Search)
- Served a self-hosted open-weights LLM (vLLM) on local GPUs and integrated it into the observatory's notebook platform (jupyter-ai)

Lead Developer, STIPS

December 2024 – Present

University of California Observatories — Lick Observatory

- First author and lead developer of [STIPS](#), the Small Telescope Image Processing Suite, an open-source pipeline that wraps the Rubin/LSST Science Pipelines for 1-m telescopes: instrument signature removal, difference imaging, coadd-template generation, and forced photometry
- Validated against published results: reproduced a Nickel supernova lightcurve and calibrated Landolt standards to better than 0.05 mag with sub-pixel astrometry
- Extended the framework to a second instrument, the CTIO 1.0-m; runs locally, in Docker, and on Slurm/HTCondor clusters

Research Assistant

May 2019 – March 2020

Department of Physics, Middlebury College

- Built Python pipelines to retrieve and process archival Hubble Space Telescope data via MAST, finding 21 candidate dual-quasar systems
- For senior thesis, modeled host galaxy morphologies of the three most promising candidates with GALFIT and TinyTim PSFs

Cataclysmic Variables Research Project

August 2024 – 2025

City College of San Francisco

- Observational project led by Professor Claia Bryja monitoring cataclysmic variable stars; collected and reduced data from the Nickel Telescope at Lick Observatory

Presentations

- Gause, D., Telkamp, Z., Plunkett, A., Mason, B., Nikutta, R., Juneau, S., Zacharia Anil, A., Loomis, R., Offner, S., Durrett, G., Torrey, P., & Murphy, E. “MANNA: MCP Architecture for NRAO and NOIRLab Archives.” Poster, CosmicAI Cosmic Horizons Conference, Charlottesville, VA, July 2026.
- Gause, D. & Westfall, K. “The Small Telescope Image Processing Suite (STIPS): Bringing the LSST Stack to 1-m Telescopes.” Poster, 248th Meeting of the American Astronomical Society, Pasadena, CA, June 2026.
- Gause, D. “On the Hunt for Dual Quasars.” Keck Northeast Astronomy Consortium Symposium, Vassar College, October 2019.
- Gause, D. “Searching for DQSOs in Archival Hubble Data.” Poster, Middlebury Summer Research Poster Session, July 2019.

Professional Experience

Full Stack AI/ML Engineer

May 2021 – September 2026

Accenture Federal Services

- Lead developer for a small team, owning architecture and code quality on production systems
- Built data and machine-learning pipelines over large geospatial datasets, including models trained on NASA satellite imagery

Volunteer and Outreach

Scientific Computing Department Volunteer

March 2024 – September 2026

California Academy of Sciences

- Built and deployed a Python package using machine-learning clustering for collections data
- Sped up a processing application by 40% through parallelization

Service Learner Volunteer

August 2024 – January 2025

City College of San Francisco

- Helped plan and host monthly public sky viewing nights and planetarium shows

Skills and Certifications

- **Languages:** Python, C++, C, SQL/ADQL, Bash
- **Astronomy:** LSST Science Pipelines, Gen3 Butler, Astropy, pyvo, FITS, IVOA protocols
- **Computing:** Slurm, HTCondor, Docker, Apptainer, HPC, CI/CD, pytest, Git
- **Certifications:** CompTIA Security+ (2025)